PARAVISION

Paravision Gen 6 **Face Recognition Model Comparison and Selection Guide**

Paravision's sixth generation face recognition software delivers world-leading accuracy, deployment flexibility, developer experience, and speed. The Gen 6 face recognition has delivered shining results in NIST's Face Recognition Technology Evaluations (FRTE, formerly FRVT), including ranking as #6 in the world for 1:1 verification and #4 in the world for 1:N identification. With four model options, Paravision Gen 6 enables improved flexibility to accommodate for any partner use case. This document explores the differences between the Paravision Gen 6 face recognition models.

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August 2023

Fast

The Fast model is the smallest and highest velocity model in Paravision's face recognition suite. It delivers the most efficient performance for use cases requiring high-speed face detection and recognition with a less than 15 ms runtime. The Fast model is the best choice for use cases where the lowest processing times, highest inter-platform compatibility and best user experiences matters, and is optimized for mobile & edge applications.

Balanced

The Balanced model brings together a valuable combination of speed and accuracy. With its low footprint and high accuracy, the Balanced model is right for any partner seeking to offer accurate face recognition with low processing times. The Balanced model is designed specifically for server side deployments where GPU support is not available.

Precision

The Precision model has been optimized for high-accuracy use cases, where low error rates are vital. With a response time of under 30 milliseconds, the Precision model offers fast detection and recognition at leading accuracy levels to enable secure identity applications and matching solutions. The Precision model has been designed for server side applications where GPU support is available, and specifically optimized for Nvidia Orin devices.

Precision+

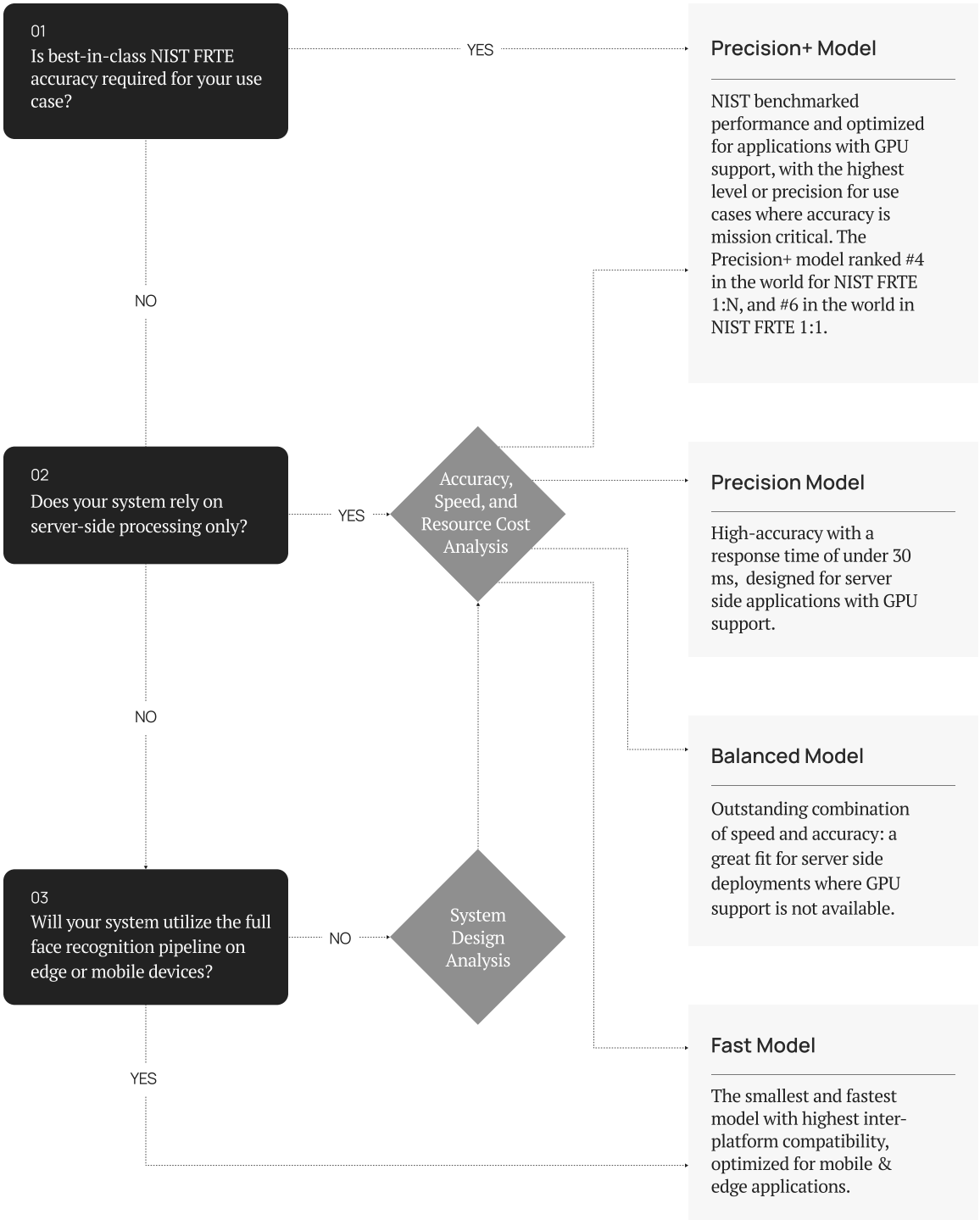
The Precision+ model is the crown jewel of Paravision's industry leading accuracy. With its NIST benchmarked performance, it offers the highest level or precision for use cases where accuracy is mission critical. The Precision+ model ranked as #4 in the world for NIST FRTE (formerly FRVT) 1:N, and #6 in the world for NIST FRTE (formerly FRVT) 1:1, achieving the most accurate rank across both tests across vendors from the U.S., UK, and EU. The Precision+ model has been designed for GPU-powered server side applications.

Technical Specifications

	Fast	Balanced	Precision	Precision+
Accuracy* @FMR=1e-6	FNMR **0.097**	FNMR **0.043** 56% more accurate than Fast	FNMR **0.033** 23% more accurate than Balanced	FNMR **0.027** 28% more accurate than Precision
Runtime**	**13.27 ms**	**17.84 ms** 34% slower than Fast	**25.61 ms** 44% slower than Balanced	**45 ms** 76% slower than Precision
Size	**80 MB**	**303 MB** 279% larger than Fast	**764 MB** 152% larger than Balanced	**1.51 GB** 97% larger than Precision
Availability	**Edge** (Ambarella), **Mobile** (iOS, Android), **Server**	**Server** CPU & GPU	**Server** CPU & GPU	**Server** CPU & GPU
Use cases	Max speed and inter-platform compatibility	Fast processing even with servers without GPU support	Server use cases with GPU support with less than 30 ms respond time	Best, NIST tested accuracy for highest precision use cases

*All accuracy metrics from Paravision internal benchmarks with wild datasets. **TensorRT on NVIDIA T4, OpenVINO also supported

Accuracy improvements: from Gen 5 to Gen 6

	Fast	Balanced	Precision	Precision+
FNMR@FMR=1e-1	-89%	-78%	-72%	NA
FNMR@FMR=1e-2	-85%	-86%	-76%	NA
FNMR@FMR=1e-3	-55%	-84%	-81%	NA
FNMR@FMR=1e-4	-33%	-69%	-71%	NA
FNMR@FMR=1e-5	-20%	-54%	-52%	NA
FNMR@FMR=1e-6	-10%	-30%	-19%	NA
Average	-48%	-67%	-62%	NA

Selecting the right Paravision model